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THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

DISCRETE MATHEMATICS AND GRAPH THEORY

Time: 3 Hours **Maximum Marks: 100**

Instructions: **(i) Missing data may be suitably assumed**

(ii) Answer all Questions

		PART-A	**5 x2=10**		
1.	**a.**	Find the number of ways of the alphabets A, B, C, D, E, F, G are arranged such that A is not in first position, B is not in second position ... G is not in seventh position.	L3	CO2	**02**
	b.	Find the sequence for the following exponential generating function $\frac{3}{1-2x} + e^x$.	L2	CO3	**02**
	c.	Solve: $a_n - 7a_{n-1} + 10a_{n-2} = 0$	L3	CO4	**02**
	d.	Define simple graph along with example.	L1	CO1	**02**
	e.	If a tree T has 3 vertices of degree 2, 2 vertices of degree 3 and 2 vertices of degree 4. Find the number of pendant vertices in T.	L2	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	State product rule. Cars of a particular manufacturer come in 4 models, 12 colors, 3 engine sizes, and 2 transmission types. **i.** How many distinct cars (of this company) can be manufactured? **ii.** Of these how many have the same colour?	L3	CO2	**05**
		OR			
	b.	Find the number of 3-digit even numbers with no repeated digits.	L3	CO2	**05**
3.	**a.**	In each of the following, $f(x)$ is a generating function for the sequence sequence $\langle a_r \rangle$ and $g(x)$ is a generating function for a sequence $\langle b_r \rangle$. Express $g(x)$ in terms of $f(x)$. $b_1 = 1, b_3 = 3, b_7 = 7, b_n = 2a_n + 5$ for $n \neq 1,3,7$.	L2	CO3	**05**
		OR			
	b.	Using generating function, find the number of non-negative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 25$.	L3	CO3	**05**
4.	**a.**	Solve the recurrence relation $a_n - 3a_{n-1} = 5 \times 3^n$ for $n \geq 1$ given that $a_0 = 2$.	L3	CO4	**05**
		OR			
	b.	Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with the initial conditions $a_0 = 1$ and $a_1 = 4$.	L3	CO4	**05**

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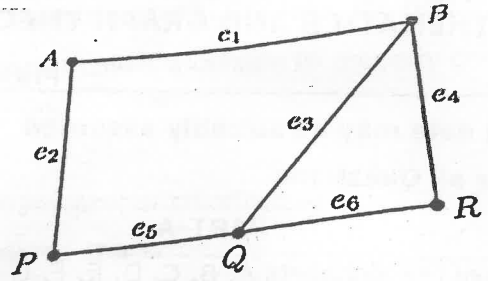
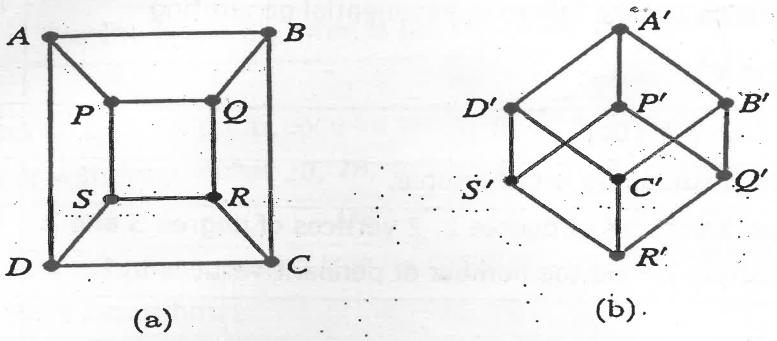
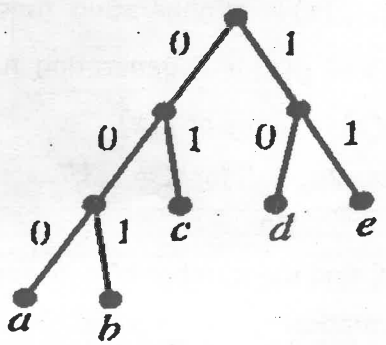
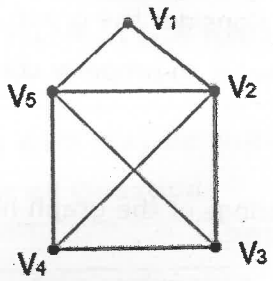
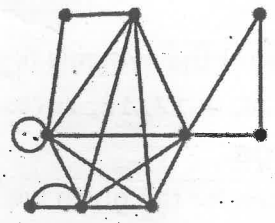
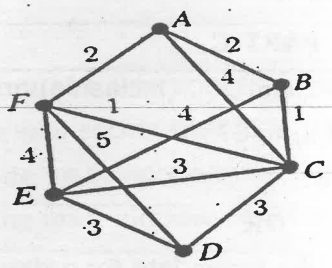
5.	a.	Consider the graph shown below. Find all paths from vertex A to vertex R. Also, indicate their lengths. 	L3	CO5	05
		OR			
	b.	Verify that the two graphs shown below are isomorphic. 	L3	CO5	05
6.	a.	A computer laboratory of a school has 10 computers that are to be connected to the wall socket that has two outlets. The connections are made by using extension cords that have two outlets each. Find the least number of cords needed to get these computer setup for use.	L3	CO6	05
		OR			
	b.	Obtain the prefix code represented by the following labeled complete binary sequence 	L3	CO6	05

7.	a.	Find the chromatic polynomial for the graph shown below using decomposition theorem. If 5 colors are available, in how many ways can the vertices of this graph be properly colored? 	L3	CO5	**05**
		OR			
	b.	Check the planarity (or otherwise) of the following graph by the method of elementary reduction. 	L3	CO5	**05**
		PART-C	**6 X 10=60**		
8.	a.	How many integers between 1 and 300 (inclusive) are i) Divisible by at least one of 5, 6, 8? ii) Divisible by none of 5, 6, 8?	L3	CO2	**10**
		OR			
	b.	Four persons P1, P2, P3, P4 who arrive late for a dinner party find that only one chair at each of five tables T1, T2, T3, T4 and T5 is vacant. P1 will not sit at T1 or T2, P2 will not sit at T2, P3 will not sit at T3 or T4, and P4 will not sit at T4 or T5. Find the number of ways they can occupy the vacant chairs.	L3	CO2	**10**
9.	a.	Using generating function, find the number of partitions of n = 6 and also write all the possibilities.	L3	CO3	**10**
		OR			
	b.	In how many ways can we distribute 24 pencils to 4 children so that each child gets at least 3 pencils but not more than eight.	L3	CO3	**10**
10.	a.	Solve the recurrence relation $a_n - a_{n-1} - 2a_{n-2} = 4$ with the initial conditions $a_0 = 1$ & $a_1 = -1$.	L3	CO4	**10**
		OR			
	b.	Find the generating function for the recurrence relation $a_{n+1} - a_n = 3^n, n \geq 0$ with $a_0 = 1$. Hence solve the relation.	L3	CO4	**10**

11.	a.	Define Multi Graph, Pendent Vertex, Walk, Path and Cycle with suitable examples.	L1	CO1	10
	b.	Define Complete bipartite graph. Consider the graph $K_{2,3}$. Let λ denote the number of colors available to properly color the vertices of this graph. Find: (i) how many proper colorings of the graph have vertices a, b colored the same. (ii) how many proper colorings of the graph have vertices a, b colored with different colors. (iii) The chromatic polynomial of the graph. (iv) For the graph $K_{2,n}$ what is the chromatic polynomial?	L3	CO5	10
12.	a.	Construct an optimal prefix code for the symbols a, o, q, u, y, z that occur with frequencies 20, 28, 4, 17, 12, 7, respectively.	L3	CO6	10
		OR			
	b.	Construct a minimal spanning tree for the given weighted graph using Prim's Algorithm.  	L3	CO6	10
13.	a.	i. Find the rook polynomial for the following board 1 2 3 4 5 6 7 ii. Find the number of positive integer solutions of the equation $x_1 + x_2 + x_3 = 17$ we have $x_1 \geq 1,\ x_2 \geq 1,\ x_3 \geq 1$.	L3	CO2	10
		OR			
	b.	i. Out of 30 students in a hostel, 15 study History, 8 study Economics and 6 study Geography. It is known that 3 students study all these subjects. Show that 7 or more students study none of these subjects. ii. How many permutations of 1, 2, 3, 4, 5, 6, 7, 8 are not derangements?	L3	CO2	10